Task-Relative Descriptive Privilege in Linear Gaussian Dynamical Systems

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Abstract

We ask whether a single description can be simultaneously optimal for all prediction tasks in a linear dynamical system. In a finite-dimensional linear Gaussian dynamical system, the optimal rank-r observer for a squared-error prediction target is uniquely determined by the top-r eigenspace of a target information matrix derived from the stationary covariance and the dynamics. Two targets whose information matrices have distinct top-r eigenspaces admit no jointly optimal observer. Global privilege holds if and only if the target family is r-*coherent*: all information matrices share a common top-r eigenspace. r-coherence requires strong structural alignment between the dynamics and the task family. Outside it, privilege is task-relative within this class. We illustrate the theorem numerically: a concrete no-privilege instance yields a global-privilege gap of $\Gamma = 0.026 > 0$, and structural coherence is confirmed to be perturbation-robust.

1 Introduction

A basic question in the study of dynamical systems is whether there is a single low-dimensional description that is optimal for every prediction task, or whether the optimal description depends on the target one seeks to predict. This paper answers that question exactly for finite-dimensional linear Gaussian dynamical systems (LGDS). The optimal rank-r observer is the top-r eigenspace of a target information matrix. Two targets whose information matrices have distinct dominant subspaces cannot be simultaneously optimised by any single observer. The exception has an exact algebraic characterisation: r-coherence of the target family.

This yields three things useful for the present question: an exact risk formula with no approximation, a proof that task-relative privilege is the rule within this class, and an explicit algebraic criterion for when global privilege is warranted.

This result is intentionally narrow. It concerns only finite-dimensional linear Gaussian dynamical systems, fixed-rank linear observers, and squared-error prediction targets.

2 System and Observer Family

Dynamics. Let

$$x_{t+1} = Ax_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, Q), \quad x_t \in \mathbb{R}^n, \tag{1}$$

with A stable (spectral radius < 1) and $Q \succ 0$. The unique stationary covariance Σ satisfies $A\Sigma A^\top + Q = \Sigma$ and is strictly positive definite.

Observer family. An observer of rank r ($1 \le r < n$) is parametrised by a row-orthonormal matrix $U \in \mathbb{R}^{r\times n}$, $UU^\top = I_r$:

$$\phi_U(x) = U\Sigma^{-1/2}x. \tag{2}$$

Pre-whitening by $\Sigma^{-1/2}$ places all observers on a common isotropic footing. Two observers differing only by an orthogonal rotation of their row spaces are equivalent; the observer space is the Grassmannian $\mathrm{Gr}(r, n)$, a smooth compact Riemannian manifold of dimension $r(n-r)$, with metric and geodesic structure from $O(n)$.

Target family. A prediction target is a pair (C, τ) with $C \in \mathbb{R}^{p\times n}$ and integer $\tau \ge 1$:

$$y_{C,\tau} = Cx_{t+\tau}. \tag{3}$$

Performance is total mean squared prediction error. A target family $\mathcal{T} = \{(C_k, \tau_k)\}$ is any finite collection.

3 Theorem

3.1 Exact Bayes risk formula

Given observation $z = U\Sigma^{-1/2}x_t$, the MMSE predictor of $Cx_{t+\tau}$ is linear by Gaussian conjugacy. At stationarity, $\mathrm{Cov}(Cx_{t+\tau}, z^\top) = CA^\tau\Sigma^{1/2}U^\top$ and $\mathrm{Cov}(z, z^\top) = I_r$. The Bayes risk is

$$\mathcal{R}(U;\, C, \tau) = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}\big(U\, M_{C,\tau}\, U^\top\big), \tag{4}$$

where the *target information matrix* is

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top C A^\tau \Sigma^{1/2} \;\in\; \mathbb{R}^{n\times n}. \tag{5}$$

$M_{C,\tau}$ is symmetric positive semidefinite of rank at most $\min(p, n)$. The formula is exact.

3.2 Top-eigenspace optimality

Proposition 1. *For fixed rank r, the observer minimising $\mathcal{R}(U;\, C, \tau)$ over $\mathrm{Gr}(r, n)$ has row space equal to the top-r eigenspace of $M_{C,\tau}$. The minimum risk is*

$$\mathcal{R}^* = \mathrm{tr}(C\Sigma C^\top) - \sum_{i=1}^{r} \lambda_i(M_{C,\tau}), \tag{6}$$

where $\lambda_1 \ge \cdots \ge \lambda_n$ are the eigenvalues of $M_{C,\tau}$. Uniqueness up to rotation holds when $\lambda_r > \lambda_{r+1}$.

Proof. Minimising $\mathcal{R}$ is equivalent to maximising $\mathrm{tr}(UM_{C,\tau}U^\top)$ subject to $UU^\top = I_r$. By the Ky Fan variational principle the maximum equals $\sum_{i=1}^{r} \lambda_i(M_{C,\tau})$, achieved exactly when the row space of U spans the corresponding eigenspace. □

3.3 No-global-privilege corollary

Corollary 2. *If targets (C_1, τ_1) and (C_2, τ_2) have information matrices M_1 and M_2 with distinct top-r eigenspaces, no single rank-r observer is simultaneously optimal for both.*

Proof. Optimality for target k requires the row space of U to equal the top-r eigenspace of M_k. Distinct eigenspaces cannot be spanned by the same U. □

The corollary extends to any finite target family: if $\{M_k\}$ do not share a common top-r eigenspace, no rank-r observer is globally optimal.

3.4 r-coherence: the exact exception

Definition 3. *A target family $\mathcal{T}$ is* r-coherent *with respect to (A, Σ) if all matrices $\{M_{C,\tau} : (C, \tau) \in \mathcal{T}\}$ share a common top-r eigenspace V^*.*

Corollary 4. *Global privilege holds if and only if $\mathcal{T}$ is r-coherent.*

r-coherence requires all targets to be linear functionals of a common r-dimensional sufficient statistic across all horizons – a strong structural alignment between the dynamics and the task family. Outside r-coherence, the optimal observer generally depends on the target within the class studied here.

Remark 5. *r-coherence is an algebraic equality between eigenspaces, not an approximate or asymptotic condition. It is the exact characterisation of when global privilege is warranted.*

4 Numerical Illustrations

The theorem is analytically exact under its stated assumptions. We illustrate it numerically with two finite-dimensional instances: a no-global-privilege example and a structurally coherent example. All reported quantities are computed exactly from the analytic Bayes risk formula and numerical optimization on the Grassmannian.

Setup. State dimension $n = 8$, observer rank $r = 2$. $A = VDV^{-1}$ with eigenvalues drawn from $U[0.3, 0.85]$, random non-orthogonal V (seed 100), $Q = 0.1\, I_8$. Σ is computed from the discrete Lyapunov equation; $\Sigma \succ 0$ is confirmed to machine precision. All reported quantities are computed exactly from the analytic Bayes risk formula, eigenspace calculations, and repeated projected optimization over $\mathrm{Gr}(2, 8)$.

Condition A: no-global-privilege instance. Targets $T_1 = ([e_1^\top; e_2^\top], 1)$ and $T_2 = ([e_5^\top; e_6^\top], 5)$. With non-normal A, A^1 and A^5 rotate state-space directions differently; the principal angles between the top-2 eigenspaces of M_1 and M_2 are $54.1°$ and $24.0°$, confirming that no single rank-2 observer is simultaneously optimal.

The global-privilege gap

$$\Gamma = \min_{U \in \mathrm{Gr}(r,n)} \max_k \left[\mathcal{R}(U; C_k, \tau_k) - \mathcal{R}_k^*\right] \tag{7}$$

measures the best achievable worst-case regret under task conflict. The minimax observer U_{joint}, found by repeated projected optimization over $\mathrm{Gr}(2, 8)$ from random initializations, achieves

Table 1: Regret matrix, Condition A ($n = 8$, $r = 2$, seed 100). All values from the exact analytic Bayes risk formula.

Observer	Regret on T_1	Regret on T_2
U_1^* (optimal for T_1)	0.000	0.036
U_2^* (optimal for T_2)	0.763	0.000
U_{joint} (compromise)	0.026	0.026

$\Gamma = 0.026 > 0$. No globally optimal observer exists for this target family. Table 1 gives the full regret structure. The asymmetry in the off-diagonal entries reflects the geometry of the instance: U_2^* is optimised for directions nearly orthogonal to T_1, so its cross-regret is large; U_1^* retains partial overlap with the long-horizon T_2 target, so its cross-regret is small. Both are computed from the same analytic formula.

Condition B′: structural coherence and its robustness. A target family is constructed so that all M_k share the same top-2 eigenspace by design: normal $A_{\text{sym}} = VDV^\top$ with orthogonal V, and a common C whose rows are the top-2 eigenvectors of A, varying only the prediction horizon $\tau \in \{1, 2, 3, 5\}$. Under this construction $M_{C,\tau}$ has the same dominant 2-dimensional eigenspace for all τ, so the four optimal observers are identical by the theorem. The maximum pairwise principal angle across all four is 0.000° to machine precision.

Under $\varepsilon = 0.1$ perturbations to C (20 random draws), the maximum principal angle across all draws and all pairs is 0.102 rad – well below the fragility threshold of 0.15 rad. Structural coherence is perturbation-robust, distinguishing it from accidental near-coincidences of eigenspaces that break under small perturbations.

The numerical results are consistent with the theorem in all cases.

5 Scope

What is earned. In finite-dimensional linear Gaussian dynamical systems at stationarity, among rank-r linear observers $\phi_U(x) = U\Sigma^{-1/2}x$ and squared-error prediction targets $y = Cx_{t+\tau}$, descriptive privilege is task-relative unless the target family is r-coherent. r-coherence requires strong structural alignment between dynamics and tasks. Outside it, no fixed-rank observer is globally optimal, and the cost of forcing any observer onto a non-native target is given exactly by the Bayes risk formula. The Grassmannian $\mathrm{Gr}(r, n)$ is the literal domain of this optimisation: a compact Riemannian manifold on which observer quality is a smooth function, with $O(n)$ acting by transformation.

What is not earned. This result does not apply to nonlinear systems, non-Gaussian noise, nonlinear observers, or non-prediction tasks. The causal axis – whether the observer optimal for intervention differs from the observer optimal for prediction – is not addressed here.